

Date	July 1, 2026
Team ID	SB-AI-2026-9284
Project Name	Sustainable Waste Management AI Assistant
Maximum Marks	2 Marks

Performance & Automated Testing Report

1. Automated API Test Coverage

A suite of 6 unit and integration tests is executed in the backend directory:

- ? test_health_check: Verifies system health status is healthy (Passed)
- ? test_classify_validation: Rejects empty or invalid strings (Passed)
- ? test_classify_fallback_logic: Resolves classifications offline (Passed)
- ? test_get_history: Asserts read/insert operations (Passed)
- ? test_get_statistics: Checks mathematical metrics calculations (Passed)
- ? test_get_centers: Filters map center coordinate listings (Passed)

2. API Latency & Load Telemetry

- Local Heuristics (100 reqs): Avg Latency 12ms, Success Rate 100%
- Groq Cloud (1 Req): Avg Latency 980ms, Success Rate 100%
- Groq Cloud (10 reqs concurrent): Avg Latency 1420ms, Success Rate 100%